www.tntextbooks.in

Let p_0 V_0 and T_0 be the pressure, Volume and temperature respectively of dry nutrogen at STP,

$$\text{Then, } \frac{P_0V_0}{T_0} = \frac{P_1V_1}{T_1}$$

$$\therefore V_0 = \frac{P_1V_1}{T_1} \times \frac{T_0}{P_0}$$

$$V_0 = \left(\frac{P_1V_1}{T_1} \times \frac{273K}{760}\right) \text{mmHg}$$

Calculation of percentage of nitrogen. 22.4 L of N_2 at STP weigh 28gof N_2

$\therefore$ V_0 L of N_2 at S.T.P weigh $\frac{28}{22.4} \times V_0$

Wg of Organic compound contain $\left(\frac{28}{22.4} \times \frac{V_0}{W}\right)$g of nitrogen

$\therefore$ Percentage of nitrogen =

$$\left(\frac{28}{22.4} \times \frac{V_0}{W}\right) \times 100$$

Problem: 0.1688 g when analyzed by the Dumas method yield 31.7 mL of moist nitrogen measured at 14^0 C and 758 mm mercury pressure. Determine the % of N in the substance (Aqueous tension at 14^0 C =12 mm)

Weight of Organic compound = 0.168g
Volume of moist nitrogen (V_1) = 31.7mL
= 31.7×10^{-3} L
Temperature (T_1) = 14^0C
= 14 +273
= 287K
Pressure of Moist nitrogen (P) = 758 mm Hg
Aqueous tension at 14^0C = 14^0C
= 12 mm of Hg
$\therefore$ Pressure of dry nitrogen = ($P-P^1$)
= 758- 12
= 746 mm of Hg

$$\frac{P_1V_1}{T_1} = \frac{P_0V_0}{T_0}$$

$$\therefore V_0 = \frac{746 \times 31.7 \times 10^{-3}}{287} \times \frac{273}{760}$$

$V_0 = 29.58 \times 10^{-3}$ L

Percentage of nitrogen $= \left(\frac{28}{22.4} \times \frac{V_0}{W}\right) \times 100$

$= \frac{28}{22.4} \times \frac{29.58 \times 10^{-3}}{0.1688} \times 100$

= 21.90%



Dumas method is a high precision method and is generally preferred over Kjeldahl's method.

Kjeldahls method:

This method is carried much more easily than the Dumas method. It is used largely in the analysis of foods and fertilizers. Kjeldahls method is based on the fact that when an organic compound containing nitrogen is heated with Conc. H_2SO_4, the nitrogen in it is quantitatively converted to ammonium sulphate. The resultant liquid is then treated with excess of alkali and then liberated ammonia gas absorbed in excess of standard acid. The amount of ammonia (and hence nitrogen) is determined by finding the amount of acid neutralized by back titration with same standard alkali.

Procedure:

A weighed quantity of the substance (0.3 to 0.5 g) is placed in a special long – necked Kjeldahl flask made of pyrex glass. About 25 mL of Conc. H_2SO_4 together with a little K_2SO_4 and $CuSO_4$ (catalyst) are added to it the flask is loosely stoppered by a glass bulb and heated gently in an inclined